Validation Architecture in delaunay

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Author TODO. Write the abstract in your own words.

Additional Key Words and Phrases: TODO author-supplied keywords

1 INTRODUCTION

- **Author TODO.** State the problem and motivation.
- **Author TODO.** Explain why validation architecture matters for a scientific computational-geometry library.
- **Author TODO.** Summarize the central contribution in your own words.
- **Author TODO.** Preview the paper structure.

2 BACKGROUND AND DEFINITIONS

- **Author TODO.** Define the triangulation, simplex, ridge, facet, and embedding terms used throughout the paper.
- **Author TODO.** Introduce PL-manifold terminology and the link condition.
- **Author TODO.** Introduce Euclidean, toroidal, and spherical embedding models.
- **Author TODO.** Introduce Delaunay and future geometric-predicate terminology.

3 VALIDATION HIERARCHY

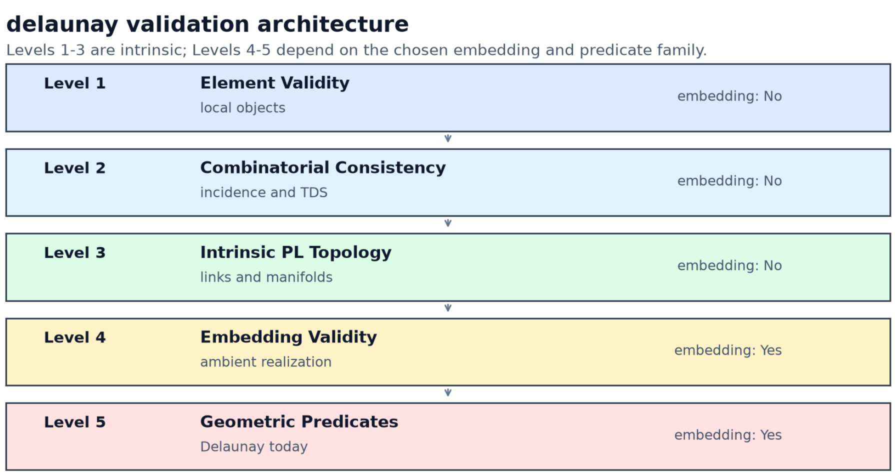

Fig. 1. **Author TODO.** Describe the validation hierarchy figure.

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3.1 Level 1: Element Validity

- **Author TODO.** Describe what individual vertices, simplices, facets, and coordinates must prove locally.
- **Author TODO.** Explain what this level deliberately does not check.

3.2 Level 2: Combinatorial Consistency

- **Author TODO.** Describe adjacency, incidence, neighbor symmetry, and TDS integrity.
- **Author TODO.** Explain the separation from topology and geometry.

3.3 Level 3: Intrinsic PL Topology

- **Author TODO.** State the PL-manifold and boundary conditions.
- **Author TODO.** Discuss pseudomanifold and strict PL-manifold policy choices.
- **Author TODO.** Explain why this layer is independent of Euclidean, toroidal, and spherical embeddings.

3.4 Level 4: Embedding Validity

- **Author TODO.** Describe faithful realization in the active ambient model.
- **Author TODO.** Compare Euclidean affine charts, toroidal periodic charts, and spherical embeddings.
- **Author TODO.** Explain why embedding validity is a precondition for geometric predicates.

3.5 Level 5: Geometric Predicates

- **Author TODO.** Describe Delaunay as the first implemented predicate family.
- **Author TODO.** Explain how Euclidean, toroidal, and spherical Delaunay predicates differ.
- **Author TODO.** Leave room for regular, weighted, constrained, Gabriel, alpha, and future predicate families.

4 IMPLEMENTATION ARCHITECTURE

- **Author TODO.** Map the five validation levels onto the Rust API surface.
- **Author TODO.** Describe the role of fast-fail checks, diagnostics, reports, and cumulative validation.
- **Author TODO.** Explain construction-time validation policy choices such as `ExplicitOnly` and `OnSuspicion`.
- **Author TODO.** Describe how spherical topology fits as a coordinate/metric backend rather than a new simplex dimension.

5 DIAGNOSTICS AND REPRODUCIBLE FIGURES

- **Author TODO.** Explain how the validation notebook generates paper figures.
- **Author TODO.** Describe representative diagnostics without duplicating API documentation.
- **Author TODO.** Connect figure generation to the reproducibility story.

6 RELATED WORK

- **Author TODO.** Place PL topology references and Pachner/Lickorish results.
- **Author TODO.** Place Euclidean robust-predicate and Delaunay references.
- **Author TODO.** Place spherical Delaunay and Riemannian-manifold references.
- **Author TODO.** Clarify which claims are architectural observations rather than literature claims.

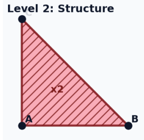
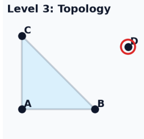
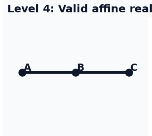
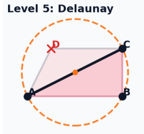
Generated failure picture	Public check / test	Explanation
Level 1: Elements 	Point::<2>::try_new src/geometry/point.rs::point_is_valid_f64	Non-finite point coordinate Element validation rejects non-finite coordinates before they can enter a vertex, simplex, or TDS. Diagnostic: Invalid coordinate at index 0 in dimension 2: NaN
Level 2: Structure 	DelaunayTriangulationBuilder::try_from_vertices_and_simplices(...).build tests/triangulation_builder.rs::test_explicit_error_variant_duplicate_simplices_structural_validation	Duplicate maximal simplex The TDS layer rejects duplicate maximal simplices because the incidence structure would no longer be a well-defined complex. Diagnostic: TDS assembly failed during explicit construction: Validation error during construction: Duplicate explicit simplices at input indices 0 and 1 with input vertex indices [0, 1, 2]
Level 3: Topology 	Triangulation::is_valid_topology tests/triangulation_builder.rs::test_explicit_unreferenced_vertices_rejected	Unreferenced vertex The topology layer rejects isolated vertices because every vertex must belong to the triangulated space. Diagnostic: Topology validation failed during explicit construction: Isolated vertex: vertex <uuid> (key VertexKey(4v1)) is not incident to any simplex
Level 4: Valid affine realization 	Triangulation::validate_embedding tests/triangulation_builder.rs::test_explicit_error_variant_geometric_nondegeneracy	Degenerate realized simplex The coordinate realization is not a valid affine realization because the abstract 2-simplex collapses to zero area. Diagnostic: Geometric nondegeneracy validation failed during explicit construction: Degenerate geometric orientation: Simplex <uuid> (key SimplexKey(1v1)) is geometrically degenerate (zero-volume simplex from collinear/coplanar vertices)
Level 5: Delaunay 	DelaunayTriangulation::is_valid_delaunay tests/triangulation_builder.rs::test_explicit_non_delaunay_mesh	Interior point in a circumcircle Point D lies inside the circumcircle of triangle ABC, so the chosen diagonal violates the local Delaunay property. Diagnostic: Delaunay validation failed during explicit construction: Delaunay verification failed: empty-circumsphere validation failed: Simplex SimplexKey(1v1) violates Delaunay property; offending vertex: Some(VertexKey(4v1))

Fig. 2. **Author TODO.** Describe the generated validation-failure examples.

7 DISCUSSION AND FUTURE WORK

- **Author TODO.** Discuss extensibility to additional embedding models and predicate families.
- **Author TODO.** Discuss limitations of the current implementation.
- **Author TODO.** Identify open validation and benchmarking questions.

8 CONCLUSION

- **Author TODO.** Write the conclusion in your own words.

ACKNOWLEDGMENTS

The author thanks Steven Carlip for discussions and guidance over the years on Causal Dynamical Triangulation, which motivated this work, and for encouragement and questions on this library, which motivated this paper.

REFERENCES

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